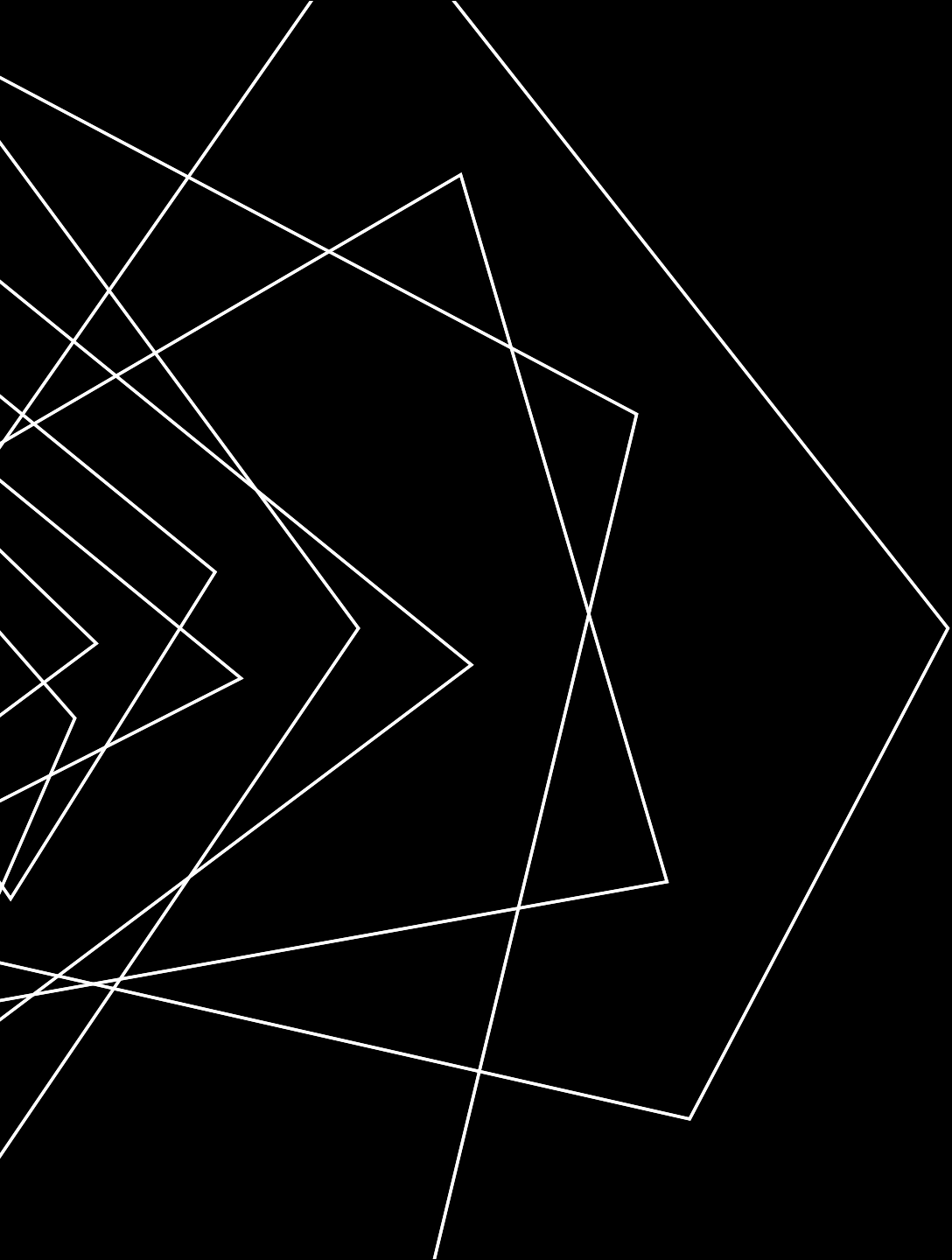




EAST TEXAS A&M
— UNIVERSITY —

CSCI 380.61E
WEB PROGRAMMING
AND INTERFACE DESIGN



API SERVERS & RESTFUL CONCEPTS

Web Programming and Interface Design -
HAPIJS

LEARNING OBJECTIVES

- Understand what an API server is.
- Learn RESTful design principles.
- Set up a Node.js + Hapi.js environment.
- Build your first REST API route.

WHAT IS AN API?

- API stands for Application Programming Interface.
- Enables communication between client and server.
- REST APIs use HTTP requests to perform CRUD operations.
- Common data format: JSON.

REST PRINCIPLES

- Representational State Transfer (REST) is a design style.
- Resources identified by URLs.
- Stateless communication (each request is independent).
- Uses HTTP verbs (GET, POST, PUT, DELETE).

HTTP METHODS OVERVIEW

Method	Description	Example
GET	Retrieve data	<code>/users</code>
POST	Create new data	<code>/users</code>
PUT	Update existing data	<code>/users/1</code>
DELETE	Delete data	<code>/users/1</code>

SETTING UP HAPI.JS ENVIRONMENT

- Install Node.js and npm.

- Initialize your project:

```
npm init -y
```

```
npm install @hapi/hapi
```

- Create an index.js file.

- Run server:

```
node index.js
```

EXAMPLE — SIMPLE HAPI SERVER

```
import Hapi from '@hapi/hapi';
```

```
const server = Hapi.server({  
  port: 3000,  
  host: 'localhost'  
});
```

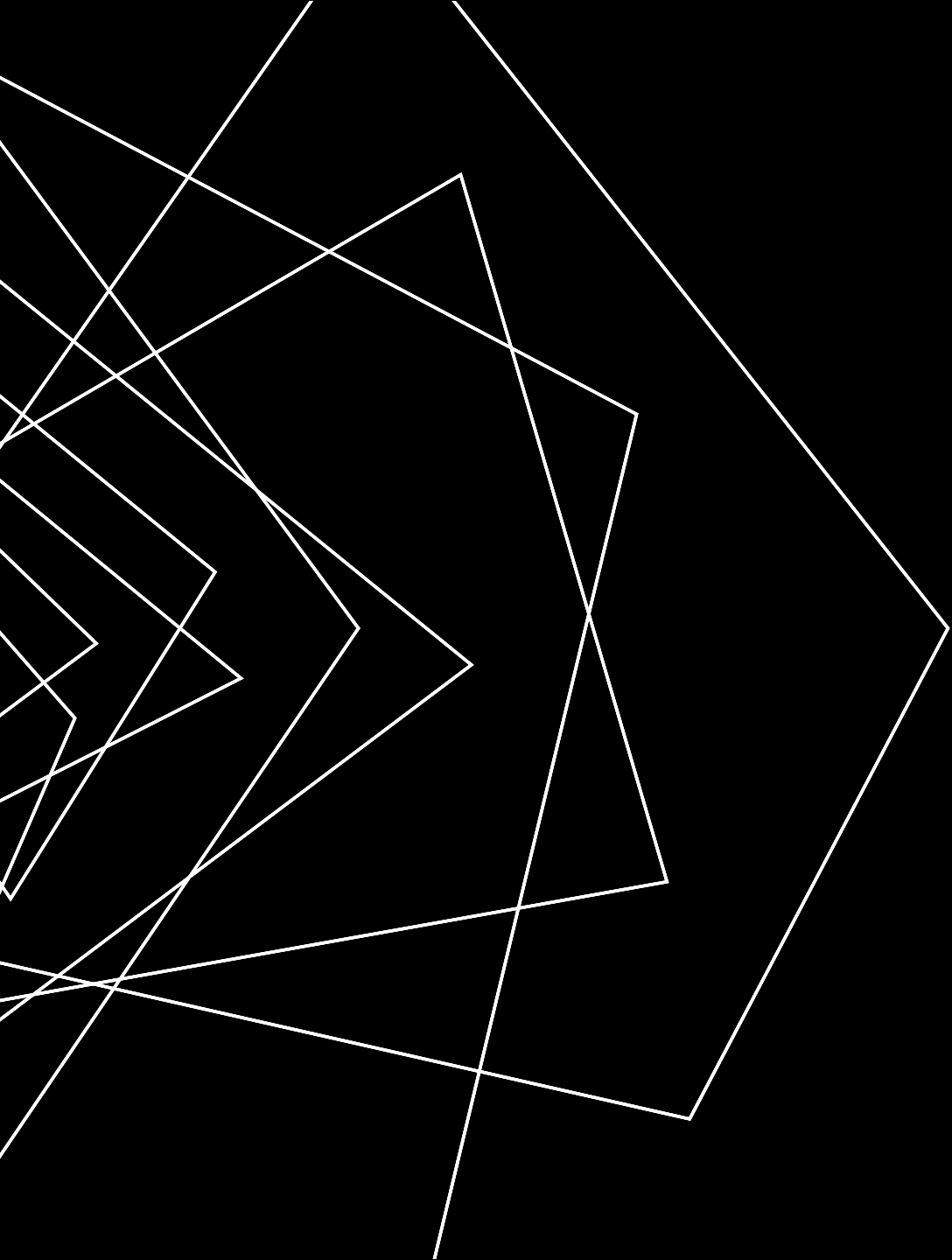
EXAMPLE — SIMPLE HAPI SERVER

```
server.route({  
  method: 'GET',  
  path: '/',  
  handler: () => 'Hello Hapi!'  
});  
  
await server.start();  
console.log('Server running on %s', server.info.uri);
```

LAB EXERCISE

- **Task:** Build your first API server
- Create routes: /hello, /time, /status.
- Each route should return JSON response.
- Example:

```
{ "status": "OK", "timestamp": "2025-10-05T20:00Z" }
```



THANK YOU

Yau Ka Cheung
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